MIL-C-48097B (AR)
30 July 1982
SUPERSEDING
MIL-C-48097A (PA)
15 July 1975

MILITARY SPECIFICATION

CHARGE, SPOTTING, PROJECTILE, CUP FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements, quality assurance provisions, and packaging for one type of cup for the projectile spotting charge.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards - Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 -	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 -	Sampling Procedures and Tables for Inspection by Attributes

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1169 - Packaging, packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications - The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9272014 - Cup

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publication - The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117 - Standard Method of Salt Spray (Fog) Testing
ASTM E 18 - Rockwell Hardness and Standard Methods of Test for Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

2.3 Order of precedence - In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Part - The part shall comply with all requirements specified on Drawing (Dwg.) 9272014, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Mechanical properties - Cold-working or heat-treating (including solution heat treatment, precipitation hardening, stress relief annealing, full annealing, etc.) operations, shall be

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performed in such a manner as to produce the specified properties uniformly throughout the entire lot. The contractor shall determine the detailed procedures to produce parts meeting the required mechanical properties. These procedures shall be documented and submitted with the first article sample to the Contracting Officer. Change in these procedures shall also be documented and submitted to the Contracting Officer.

3.4 First article inspection - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections - The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Cup	9272014	50

4.3.2 Inspections to be performed - See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection - See MIL-A-48078

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TABLE 1 - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Cup				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup (Dwg. 9272014)				
	Examination for defects	50		3.2	4.4.2.1
	Maximum corner radius at intersection of second smallest inside diameter with base	15		3.2	4.5.1
	Hardness test	15		3.2	4.5.2
	Salt spray	10		3.2	4.5.3

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078.

a. Sampling plans - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Cup				9272014
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread		0.40%	3.2	Gage
102	Minor diameter of thread		0.40%	3.2	Gage
103	Largest outside diameter		0.40%	3.2	Gage
104	True position of largest outside diameter with pitch diameter of thread		0.40%	3.2	Gage
105	Inside diameter		0.40%	3.2	Gage
106	Length of inside diameter		0.40%	3.2	Gage
107	Small outside diameter		0.40%	3.2	Gage
108	Diameter of undercut		0.40%	3.2	Gage
109	Depth of undercut		0.40%	3.2	Gage
110	Depth from face, threaded end, to bottom of undercut		0.40%	3.2	Gage
111	Runout of inside diameter to pitch diameter of thread including perpendicularity		0.40%	3.2	Gage
112	True position of small outside diameter with pitch diameter of thread		0.40%	3.2	Gage
113	Threads damaged		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9272014
4.4.2.1	Cup				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114	Cup damaged (cracked or deformed)		0.40%	3.2	Visual
Minor					
201	Total length		0.65%	3.2	Gage
202	Diameter of hole in bottom of cup		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.5	Visual

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4.4.3 Testing

4.4.3.1 Tool control of maximum corner radius at intersection of second smallest inside diameter with base (Dwg. 9272014) major defect - At the start of production, after every four (4) hours of production, or portion thereof, and whenever a change in tooling is made during production, five (5) consecutively produced cups shall be taken from each tool producing this characteristic. Each cup shall be inspected for the maximum corner radius at the intersection of the second smallest inside diameter with the base. The method of inspection shall be as specified in 4.5.1. If any cup is found to be defective by this inspection, the tool which produced this defect shall be immediately removed from production. Furthermore, the cups produced by the defective tool since the last satisfactory inspection shall be rejected until they have been inspected to the degree necessary to determine compliance or non-compliance with the requirement for this characteristic.

All tools removed from production because of some fault, may, after reworking, be returned to production providing five (5) cups consecutively produced by this tool successfully pass this inspection. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.3.2 Hardness test (Dwg. 9272014), major defect - Ten (10) cups shall be selected from each heat treatment batch and tested as specified in 4.5.2. If any cup fails to comply with the applicable requirement, the heat treatment batch represented by the sample shall be rejected.

Note 1: A heat treatment batch is herein defined as cups, of the same type, that have been produced from one lot of material, and heat treated at the same time in the same furnace, quench medium and aging process for each of all phases of the entire heat treatment process. When a continuous-type heat treatment process is used, the quantity of cups processed through the furnace for each four (4) hours production, or portion thereof shall be considered as a heat treatment batch.

4.4.3.3 Salt spray (Dwg. 9272014) major defect - Ten (10) cups shall be selected from each lot. A lot shall consist of a day's production or 1000 parts which ever is greater. If any cup fails to comply with the salt spray requirements specified on the drawing, the lot, as referenced, shall be rejected. Testing shall be in accordance with paragraph 4.5.3.

4.4.4 Inspection equipment - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inpsection Method column in the tables starting with Paragrah 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

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4.5 Test methods and procedures -

4.5.1 Inspection of maximum corner radius at intersection of second smallest inside diameter with base - The method of inspection utilized shall be non-destructive. An impression of a portion (less than half) of the inside circumference of the required surface shall be projected and inspected. The impression material shall have a shrinkage rate of no more than 2.0 percent. (Non-destructive test).

4.5.2 Hardness test - The cups shall be subjected to the hardness test in accordance with ASTM E 18. Three (3) readings shall be taken on each cup on the outside surface of the smallest diameter, approximately 120° apart, and half way between the edge of the outside surface and the hole. A suitable fixture shall be utilized to provide support under the surface being tested. If the average value of the three readings for any cup is less than the value specified on the applicable drawing, that cup shall be classed defective.

4.5.3 Salt spray - The cups shall be subjected to the salt spray test in accordance with ASTM B177. Salt solution shall be 5 percent. This test is a destructive test. Parts so tested shall not be returned to the lot.

5. PACKAGING

5.1 Preservation

5.1.1 Level C - The cup shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing -

5.2.1 Level C - The cups preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking - Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping - When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use - The component covered by this specification is intended for use with the Charge, Projectile, Spotting.

6.2 Ordering data - See MIL-A-48078.

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6.3 Submission of inspection equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to Commander, US Army ARRADCOM, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Drawings - Drawings listed in Section 2 of this specification under the headings U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Changes from previous issue - The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of ths document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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